

# **Know Thyself: Free Credit Reports and The Retail Mortgage Market**

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## Introduction

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- Top reason for mortgage rejection: Consumers' **credit history**.
  - ▶ 28% of all rejections, or 39.4% of all rejections with stated reasons.
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- **Discouraged borrowers**: Why ask, I will be rejected anyway!
  - ▶ Consumers: About 15% of the U.S. households (SCF 1998-2007).
  - ▶ UK SME Firms: Twice of those who faced a rejection (Freel et al. 2012).
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## Research Question

Examine the causal link between cheaper credit reports to consumers and **mortgage demand** and **approval ratio**.

- Information disclosure and credit market outcomes:
  - ▶ Disclosure to consumers: [Homonoff et al.\(2019\)](#), [Kulkarni et al.\(2018\)](#), [Kronlund et al.\(2019\)](#).
  - ▶ Disclosure to lenders: [Lieberman et al.\(2018\)](#), [Dobbie et al.\(2016\)](#).
  - ▶ This paper shows that the reduced economic cost of accessing one's own credit report raises the mortgage demand and approval ratio.
- Financial literacy and household behavior:
  - ▶ [Gerardi et al.\(2010\)](#), [Lusardi et al.\(2009\)](#), [Perry\(2008\)](#), [Courchane et al.\(2008\)](#), [Hundtofte\(2017\)](#).
  - ▶ This paper shows that the approval ratio and demand increased more in higher educated population.
- Mortgage Expansion in the pre-2007 crisis period:
  - ▶ [Mian et al.\(2009\)](#), [Adelino et al.\(2016\)](#), [Foote et al.\(2016\)](#), [Keys et al.\(2010\)](#), [Hundtofte\(2017\)](#).
  - ▶ Here, I show a new consumer demand channel of mortgage expansion.
- Issues related to credit reports:
  - ▶ [Avery et al.\(2003\)](#), [Nott et al.\(2003\)](#), [Cassady et al.\(2004\)](#).
  - ▶ Here, the focus is on the economic cost of credit reports.



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Consumer self-learning mechanism:

- Credit reports aid in self-assessment of creditworthiness.
- Consumers may take steps: Delay application, correct credit history, search for a more suitable lender (subprime lender), or don't apply at all.

## How do cheaper credit reports affect mortgage demand and approval ratio?

Consumer self-learning mechanism:

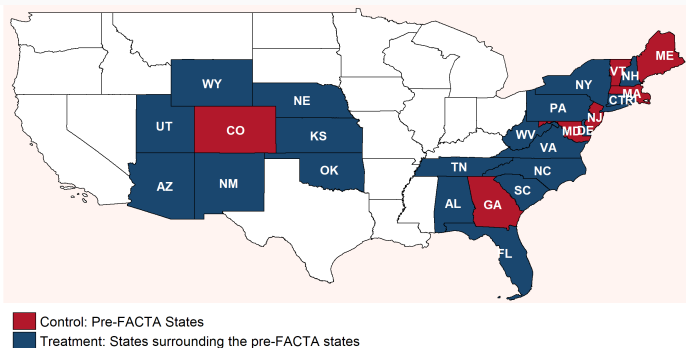
- Credit reports aid in self-assessment of creditworthiness.
- Consumers may take steps: Delay application, correct credit history, search for a more suitable lender (subprime lender), or don't apply at all.
- Effect on outcome variables:
  - ▶ Better informed consumers  $\Rightarrow$  Better pool  $\Rightarrow$  **Increase in approval ratio.**
  - ▶ Demand for credit: May increase or decrease depending on *ex-ante* proportion of consumers who underestimate their creditworthiness.



- Enactment of the federal *Fair and Accurate Transactions Act of 2003* (FACTA) on December 4, 2003.
- FACTA grants every consumers the right to obtain a free credit report from each of the nationwide consumer reporting agencies annually.
- **Event:** Establishment of [www.annualcreditreport.com](http://www.annualcreditreport.com) in Jan 2005 to distribute Free Credit Reports.
- Seven U.S. states **already** had free credit report laws (pre-FACTA states).
- I use a Difference-in-differences (DID) setting:
  - ▶ Control states: The seven Pre-FACTA states.
  - ▶ Treated states: States surrounding the Pre-FACTA states.

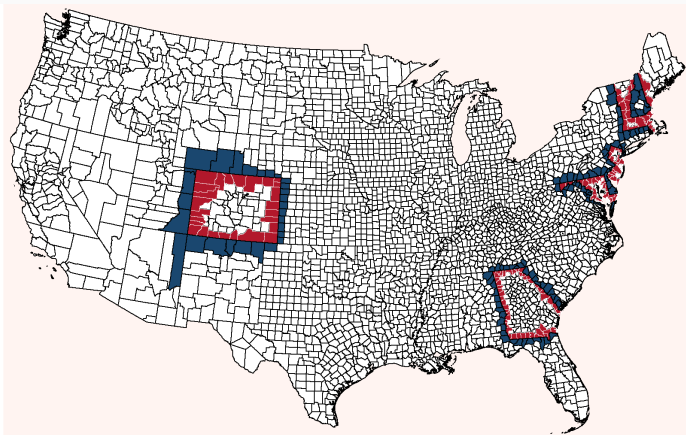
# Natural Experiment: Illustration

## Empirical Setting:



# Natural Experiment: Illustration

**Empirical Setting:** Focus only on the bordering counties



Control: Counties at the border of pre-FACTA States

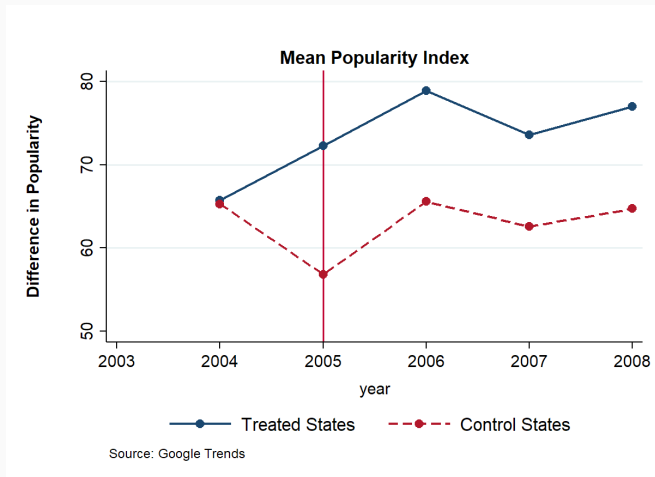


Treatment: Counties which belong to the bordering states and are adjacent to the control counties



## Natural Experiment: Changes in Interest in Credit Reports

We see that the search term **“Free Credit Report”** becomes more popular in treated areas after the experiment:



# Data and Summary Statistics

- Home Mortgage Disclosure Act (HMDA) application-level data.
- Fannie Mae and Freddie Mac mortgage application-level data.
- Equifax data on county-level subprime population.
- Census 2000 data and Annual County Business Pattern data.

## Summary Statistics

|                           | Full Sample |       |       |       | Control Group (C) |       |       |       | Treatment Group (T) |       |       |       |
|---------------------------|-------------|-------|-------|-------|-------------------|-------|-------|-------|---------------------|-------|-------|-------|
|                           | N           | Mean  | SD    | Med.  | N                 | Mean  | SD    | Med.  | N                   | Mean  | SD    | Med.  |
| Num of App per 1000 adult | 86817       | 83.42 | 74.76 | 66.41 | 36494             | 98.42 | 77.58 | 77.74 | 50323               | 72.54 | 70.68 | 56.48 |
| Approval Ratio            | 82713       | 0.63  | 0.13  | 0.64  | 35879             | 0.65  | 0.12  | 0.66  | 46834               | 0.61  | 0.13  | 0.62  |
| Deny Credit Hist Ratio    | 82713       | 0.06  | 0.04  | 0.05  | 35879             | 0.05  | 0.04  | 0.04  | 46834               | 0.06  | 0.05  | 0.05  |
| Deny Debt-to-inc Ratio    | 82713       | 0.03  | 0.03  | 0.03  | 35879             | 0.03  | 0.02  | 0.03  | 46834               | 0.03  | 0.03  | 0.03  |
| Withdrawn Ratio           | 82713       | 0.12  | 0.05  | 0.12  | 35879             | 0.12  | 0.04  | 0.11  | 46834               | 0.12  | 0.06  | 0.12  |
| $\Delta$ Inc per capita   | 2295        | 0.04  | 0.06  | 0.04  | 1143              | 0.04  | 0.05  | 0.04  | 1152                | 0.05  | 0.07  | 0.04  |
| $\Delta$ Emp              | 2298        | 0.01  | 0.09  | 0.01  | 1138              | 0.01  | 0.09  | 0.01  | 1160                | 0.01  | 0.09  | 0.01  |
| $\Delta$ State GDP        | 81          | 0.04  | 0.02  | 0.05  | 31                | 0.05  | 0.02  | 0.04  | 50                  | 0.04  | 0.03  | 0.05  |

## Results

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## Result I

$$Y_{icsjt} = \beta_0 + \beta_1 \text{Treatment}_{icsj} \times \text{Post}_t + \delta \times \text{Economic controls} + \alpha_i + \gamma_{jt} + \epsilon_{it}$$

### Baseline Result

|                         | (1)                | (2)                | (3)               | (4)               |
|-------------------------|--------------------|--------------------|-------------------|-------------------|
|                         | # Applications     | # Applications     | Approval Ratio    | Approval Ratio    |
| Treat $\times$ Post     | 13.29***<br>(2.94) | 15.45***<br>(3.65) | 0.01***<br>(2.73) | 0.01***<br>(2.97) |
| Economic Controls       | No                 | Yes                | No                | Yes               |
| Census Tract FE         | Yes                | Yes                | Yes               | Yes               |
| Border $\times$ Year FE | Yes                | Yes                | Yes               | Yes               |
| Cluster (County)        | Yes                | Yes                | Yes               | Yes               |
| R <sup>2</sup> (Adj.)   | 0.807              | 0.808              | 0.739             | 0.735             |
| Observations            | 86807              | 84786              | 82667             | 80665             |

- Mortgage applications  $\uparrow$  by 13.8—16.0% (\$38.1 billion).
  - More consumers learned that they are creditworthy.
- Approval ratio  $\uparrow$  by 1 pp (~\$2.8 billion).
  - Improved borrower pool.

### Takeaway:

- Good borrowers select-in. Bad borrowers exit/search suitable lenders.  
 $\Rightarrow$  Increase in mortgage demand and improvement in borrower pool.

## Result II: Treatment Effect by Year

### Baseline Result: Treatment by Year

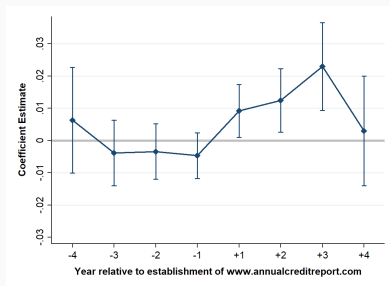


Figure 1. Coefficient Estimates

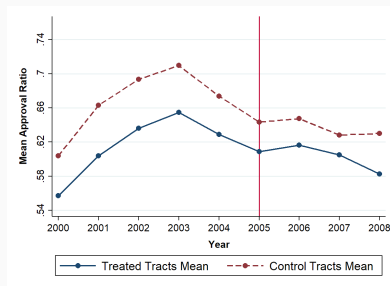


Figure 2. Raw Plot

## Result III: Improvement in Borrower Pool: Subprime Population

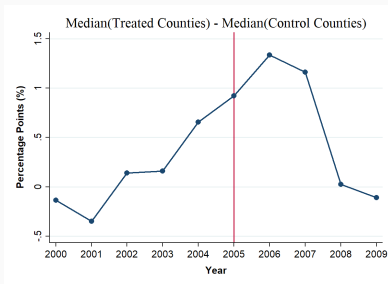


Figure 3. Median Subprime Population %

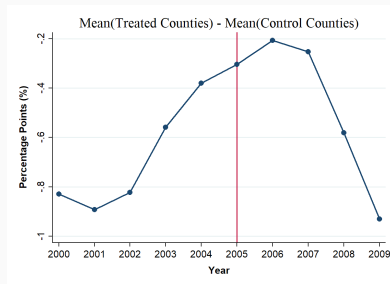


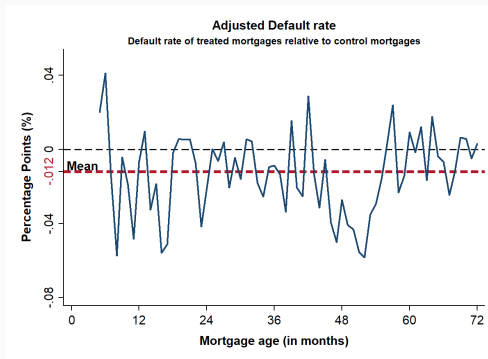
Figure 4. Mean Subprime Population %

$$\text{Median Difference} = \text{Median}(\text{Subprime \%})_{Treated} - \text{Median}(\text{Subprime \%})_{Control}$$

### Takeaways:

- ↓ Sub-prime population %  $\Rightarrow$  Improved borrower pool.
- Treated areas saw improvement in borrower pool.

## Result IV: Mortgage Default



$$\text{Adjusted default rate}_{age} = (\text{Def}_{2005,age} - \text{Def}_{2004,age})_{treated} - (\text{Def}_{2005,age} - \text{Def}_{2004,age})_{control}$$

where *Age*: months since origination, *Def*: Fraction of total mortgages of vintage year that defaults in a given month.

**Takeaway:** Lower *ex-post* default suggests improved borrower pool.

- Results suggestive of self-learning Mechanism:
  - ▶ Credit-history induced rejections ↓.
  - ▶ First-time homebuyers ↑.
  - ▶ In-process application withdrawals ↓  $\equiv$  *ex-ante* search accuracy ↑.
- A demand-driven mechanism:
  - ▶ Interest rates ↑.
  - ▶ Heterogeneity tests: Higher approval ratio in high creditworthiness and high education areas, and among low income quartile consumers.
  - ▶ No difference between areas with high- and low- lender density.



- Cheaper credit reports  $\Rightarrow$  Increase in mortgage demand and improved borrower pool.
- Consistent with increase in consumer self-learning.
- Mortgage market based findings, but apply to **any credit-related consumer decision-making**.

# Thank You

*Any suggestions, questions, or comments are welcome.*

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## *Ex-ante* Take-up Rate

For the experiment to be valid, we need to ensure that:

- Usage of the credit report **should be higher** in control states relative to the treated states in the pre-experiment period:
  - ▶ Usage of credit report in following pre-FACTA states was higher than the national average by:
    - GA 250%
    - MD 204%
    - CO 153%
    - NJ 35%
    - MA 25%
- Credit environment was better in the control states (pre-FACTA states):
  - ▶ Bankruptcy: Vermont in 2002 had lowest consumer bankruptcy, Massachusetts second lowest
  - ▶ Interest rate on conventional mortgage: Vermont and Massachusetts is below county median

# Owner-Occupied Mortgages

Baseline Results: For the sub-sample of owner-occupied mortgages only

|                         | (1)               | (2)                | (3)               | (4)               |
|-------------------------|-------------------|--------------------|-------------------|-------------------|
|                         | # Applications    | # Applications     | Approval Ratio    | Approval Ratio    |
| Treat $\times$ Post     | 9.81***<br>(2.83) | 11.50***<br>(3.48) | 0.01***<br>(2.89) | 0.01***<br>(3.04) |
| Economic Controls       | No                | Yes                | No                | Yes               |
| Census Tract FE         | Yes               | Yes                | Yes               | Yes               |
| Border $\times$ Year FE | Yes               | Yes                | Yes               | Yes               |
| Cluster (County)        | Yes               | Yes                | Yes               | Yes               |
| R <sup>2</sup> (Adj.)   | 0.640             | 0.645              | 0.618             | 0.613             |
| Observations            | 86807             | 84786              | 84190             | 82256             |

## Fraction of Non-Owner-Occupied Mortgages

Fraction of total (successful) applications which are non-owner-occupied in columns 1 and 2 (3 and 4)

|                         | Fraction of total app. |                 | Fraction of successful app. |                 |
|-------------------------|------------------------|-----------------|-----------------------------|-----------------|
|                         | (1)<br>Non-ocp.        | (2)<br>Non-ocp. | (3)<br>Non-ocp.             | (4)<br>Non-ocp. |
| Treat $\times$ Post     | 0.01<br>(1.55)         | 0.01<br>(1.60)  | 0.01<br>(1.58)              | 0.01*<br>(1.69) |
| Economic Controls       | No                     | Yes             | No                          | Yes             |
| Census Tract FE         | Yes                    | Yes             | Yes                         | Yes             |
| Border $\times$ Year FE | Yes                    | Yes             | Yes                         | Yes             |
| Cluster (County)        | Yes                    | Yes             | Yes                         | Yes             |
| R <sup>2</sup> (Adj.)   | 0.129                  | 0.127           | 0.105                       | 0.104           |
| Observations            | 82667                  | 80665           | 82626                       | 80624           |

# Mechanism: Contraction in credit-history induced rejections

## Contraction in credit-history induced rejections

|                         | All Areas         |                   | High Denial Areas   |                    | All Areas         |                   | High Denial Areas |                   |
|-------------------------|-------------------|-------------------|---------------------|--------------------|-------------------|-------------------|-------------------|-------------------|
|                         | (1)               | (2)               | (3)                 | (4)                | (5)               | (6)               | (7)               | (8)               |
|                         | % C.Hist          | % C.Hist          | % C.Hist            | % C.Hist           | % DTI             | % DTI             | % DTI             | % DTI             |
| Treat $\times$ Post     | -0.003<br>(-1.50) | -0.003<br>(-1.51) | -0.003**<br>(-2.04) | -0.003*<br>(-1.88) | -0.002<br>(-1.07) | -0.001<br>(-0.96) | -0.002<br>(-1.48) | -0.002<br>(-1.19) |
| Economic Controls       | No                | Yes               | No                  | Yes                | No                | Yes               | No                | Yes               |
| Census Tract FE         | Yes               | Yes               | Yes                 | Yes                | Yes               | Yes               | Yes               | Yes               |
| Border $\times$ Year FE | Yes               | Yes               | Yes                 | Yes                | Yes               | Yes               | Yes               | Yes               |
| Cluster (County)        | Yes               | Yes               | Yes                 | Yes                | Yes               | Yes               | Yes               | Yes               |
| R <sup>2</sup> (Adj.)   | 0.542             | 0.538             | 0.575               | 0.575              | 0.267             | 0.266             | 0.319             | 0.322             |
| Observations            | 82667             | 80665             | 39071               | 38692              | 82667             | 80665             | 39071             | 38692             |

**Takeaway:** Drop in rejection due to credit history induced rejections is consistent with consumers learning their creditworthiness.

## Mechanism: Improvement in search accuracy (Drop in in-process withdrawals)

### Drop in in-process Withdrawals

|                         | (1)                     | (2)                     |
|-------------------------|-------------------------|-------------------------|
|                         | % Application Withdrawn | % Application Withdrawn |
| Treat $\times$ Post     | -0.009***<br>(-2.81)    | -0.011***<br>(-3.51)    |
| Economic Controls       | No                      | Yes                     |
| Census Tract FE         | Yes                     | Yes                     |
| Border $\times$ Year FE | Yes                     | Yes                     |
| Cluster (County)        | Yes                     | Yes                     |
| R <sup>2</sup> (Adj.)   | 0.340                   | 0.341                   |
| Observations            | 82667                   | 80665                   |

**Takeaway:** Drop in in-process application withdrawal fraction is consistent with increase in better informed consumers.

# Mechanism: First-time Homebuyers

## Fraction of First-time Homebuyers

|                        | Denominator - Applications with Known Status |                     | Denominator - All Applications |                     |
|------------------------|----------------------------------------------|---------------------|--------------------------------|---------------------|
|                        | (1)<br>% First-time                          | (2)<br>% First-time | (3)<br>% First-time            | (4)<br>% First-time |
| Treat $\times$ Post    | 0.01**<br>(2.55)                             | 0.01**<br>(2.31)    | 0.01**<br>(2.00)               | 0.01*<br>(1.78)     |
| Economic Controls      | No                                           | Yes                 | No                             | Yes                 |
| Zip3-State FE          | Yes                                          | Yes                 | Yes                            | Yes                 |
| Border $\times$ Qtr FE | Yes                                          | Yes                 | Yes                            | Yes                 |
| Cluster Zip3-State     | Yes                                          | Yes                 | Yes                            | Yes                 |
| R <sup>2</sup> (Adj.)  | 0.691                                        | 0.692               | 0.691                          | 0.691               |
| Observations           | 7706                                         | 7706                | 7711                           | 7711                |

**Takeaway:** Fraction of first-time homebuyers increases. In terms of the mechanism, this is consistent with new borrowers entering into market.



## Interest rate of the mortgages

|                        | Interest Rate (in percentage point) |        |        |        |
|------------------------|-------------------------------------|--------|--------|--------|
|                        | (1)                                 | (2)    | (3)    | (4)    |
|                        | Mean                                | Mean   | Median | Median |
| Treat $\times$ Post    | 0.012*                              | 0.011* | 0.011* | 0.010* |
|                        | (1.93)                              | (1.77) | (1.93) | (1.72) |
| Economic Controls      | No                                  | Yes    | No     | Yes    |
| Zip3-State FE          | Yes                                 | Yes    | Yes    | Yes    |
| Border $\times$ Qtr FE | Yes                                 | Yes    | Yes    | Yes    |
| Cluster Zip3-State     | Yes                                 | Yes    | Yes    | Yes    |
| R <sup>2</sup> (Adj.)  | 0.991                               | 0.991  | 0.989  | 0.989  |
| Observations           | 7711                                | 7711   | 7711   | 7711   |

**Takeaway:** Both the price and quantity of the mortgages increased, **confirming an outward shift in demand-curve.**

# Heterogeneity: Density of lenders

## Heterogeneous Effect by Lender Density

|                         | Volume (in 1000 USD) per Adult |                 |                    |                  | Approval Ratio     |                  |                    |                   |
|-------------------------|--------------------------------|-----------------|--------------------|------------------|--------------------|------------------|--------------------|-------------------|
|                         | (1)                            | (2)             | (3)                | (4)              | (5)                | (6)              | (7)                | (8)               |
|                         | Low                            | High            | Low                | High             | Low                | High             | Low                | High              |
| Treat $\times$ Post     | 0.002**<br>(2.16)              | 0.001<br>(1.11) | 0.003***<br>(2.84) | 0.002*<br>(1.78) | 0.015***<br>(2.98) | 0.009*<br>(1.90) | 0.015***<br>(3.21) | 0.010**<br>(2.03) |
| Difference [High - Low] |                                | -0.001          |                    | -0.001           |                    | -0.006           |                    | -0.006            |
| p-value                 |                                | (0.577)         |                    | (0.581)          |                    | (0.501)          |                    | (0.475)           |
| Economic Controls       | No                             | No              | Yes                | Yes              | No                 | No               | Yes                | Yes               |
| Census Tract FE         | Yes                            | Yes             | Yes                | Yes              | Yes                | Yes              | Yes                | Yes               |
| Border $\times$ Year FE | Yes                            | Yes             | Yes                | Yes              | Yes                | Yes              | Yes                | Yes               |
| Cluster (County)        | Yes                            | Yes             | Yes                | Yes              | Yes                | Yes              | Yes                | Yes               |
| R <sup>2</sup> (Adj.)   | 0.663                          | 0.595           | 0.652              | 0.589            | 0.750              | 0.716            | 0.746              | 0.712             |
| Observations            | 60699                          | 25805           | 59206              | 25289            | 57623              | 24936            | 56143              | 24426             |

**Takeaway:** There is no difference in mortgage origination or the approval ratios in areas with high- and low-lenders density.

# Heterogeneity: Education

## Heterogeneous Effect by Education

|                         | Low Education Area |                  |                   |                   | High Education Area |                    |                   |                   |
|-------------------------|--------------------|------------------|-------------------|-------------------|---------------------|--------------------|-------------------|-------------------|
|                         | (1)<br># App.      | (2)<br># App.    | (3)<br>Apv. Ratio | (4)<br>Apv. Ratio | (5)<br># App.       | (6)<br># App.      | (7)<br>Apv. Ratio | (8)<br>Apv. Ratio |
| Treat $\times$ Post     | 6.03*<br>(1.95)    | 6.88**<br>(2.20) | 0.01<br>(1.55)    | 0.01*<br>(1.66)   | 12.52***<br>(2.67)  | 15.04***<br>(3.55) | 0.01***<br>(2.92) | 0.01***<br>(3.08) |
| Economic Controls       | No                 | Yes              | No                | Yes               | No                  | Yes                | No                | Yes               |
| Census Tract FE         | Yes                | Yes              | Yes               | Yes               | Yes                 | Yes                | Yes               | Yes               |
| Border $\times$ Year FE | Yes                | Yes              | Yes               | Yes               | Yes                 | Yes                | Yes               | Yes               |
| Cluster (County)        | Yes                | Yes              | Yes               | Yes               | Yes                 | Yes                | Yes               | Yes               |
| R <sup>2</sup> (Adj.)   | 0.828              | 0.831            | 0.582             | 0.575             | 0.824               | 0.823              | 0.743             | 0.739             |
| Observations            | 35945              | 35419            | 33182             | 32666             | 50720               | 49261              | 49363             | 47913             |

**Takeaway:** Approval ratio increased significantly in high education areas.

# Heterogeneity: Creditworthiness I

## Heterogeneous Effect by Creditworthiness (County subprime population measure)

|                         | Prime Counties    |                   |                   |                   | Subprime Counties |                   |                   |                   |
|-------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                         | (1)<br># App.     | (2)<br># App.     | (3)<br>Apv. Ratio | (4)<br>Apv. Ratio | (5)<br># App.     | (6)<br># App.     | (7)<br>Apv. Ratio | (8)<br>Apv. Ratio |
| Treat $\times$ Post     | 16.73**<br>(2.29) | 18.11**<br>(2.59) | 0.02***<br>(3.14) | 0.02***<br>(3.32) | 8.49<br>(1.64)    | 10.69**<br>(2.25) | 0.01*<br>(1.70)   | 0.01**<br>(2.01)  |
| Economic Controls       | No                | Yes               | No                | Yes               | No                | Yes               | No                | Yes               |
| Census Tract FE         | Yes               | Yes               | Yes               | Yes               | Yes               | Yes               | Yes               | Yes               |
| Border $\times$ Year FE | Yes               | Yes               | Yes               | Yes               | Yes               | Yes               | Yes               | Yes               |
| Cluster (County)        | Yes               | Yes               | Yes               | Yes               | Yes               | Yes               | Yes               | Yes               |
| R <sup>2</sup> (Adj.)   | 0.802             | 0.802             | 0.777             | 0.773             | 0.826             | 0.828             | 0.681             | 0.680             |
| Observations            | 39263             | 37698             | 38184             | 36631             | 47241             | 46797             | 44375             | 43938             |

**Takeaway:** Approval ratio increased significantly in high creditworthiness areas.

# Heterogeneity: Creditworthiness II

## Heterogeneous Effect by Creditworthiness (Payday lender based measure)

|                         | # Payday Lenders - Low |                    |                   |                   | # Payday Lenders - High |                    |                   |                   |
|-------------------------|------------------------|--------------------|-------------------|-------------------|-------------------------|--------------------|-------------------|-------------------|
|                         | (1)<br># App.          | (2)<br># App.      | (3)<br>Apv. Ratio | (4)<br>Apv. Ratio | (5)<br># App.           | (6)<br># App.      | (7)<br>Apv. Ratio | (8)<br>Apv. Ratio |
| Treat $\times$ Post     | 68.43***<br>(5.31)     | 72.78***<br>(5.36) | 0.05***<br>(3.46) | 0.05***<br>(3.43) | 43.72***<br>(3.82)      | 43.27***<br>(4.00) | 0.01<br>(0.60)    | 0.01<br>(0.64)    |
| Economic Controls       | No                     | Yes                | No                | Yes               | No                      | Yes                | No                | Yes               |
| Census Tract FE         | Yes                    | Yes                | Yes               | Yes               | Yes                     | Yes                | Yes               | Yes               |
| Border $\times$ Year FE | Yes                    | Yes                | Yes               | Yes               | Yes                     | Yes                | Yes               | Yes               |
| Cluster (County)        | Yes                    | Yes                | Yes               | Yes               | Yes                     | Yes                | Yes               | Yes               |
| R <sup>2</sup> (Adj.)   | 0.790                  | 0.793              | 0.751             | 0.751             | 0.816                   | 0.818              | 0.773             | 0.774             |
| Observations            | 1452                   | 1452               | 1395              | 1395              | 872                     | 872                | 865               | 865               |

**Takeaway:** Approval ratio increased significantly in high creditworthiness areas.

# Heterogeneity: Consumer Income

## Heterogeneous Effect by Consumer Income

|                         | Income Quartile 1 |                   | Income quartile 2 |                | Income Quartile 3 |                | Income quartile 4 |                  |
|-------------------------|-------------------|-------------------|-------------------|----------------|-------------------|----------------|-------------------|------------------|
|                         | (1)<br>Aprv.      | (2)<br>Aprv.      | (3)<br>Aprv.      | (4)<br>Aprv.   | (5)<br>Aprv.      | (6)<br>Aprv.   | (7)<br>Aprv.      | (8)<br>Aprv.     |
| Treat $\times$ Post     | 0.01***<br>(2.99) | 0.02***<br>(3.97) | 0.01<br>(1.03)    | 0.01<br>(1.39) | 0.01<br>(1.13)    | 0.01<br>(1.23) | 0.01*<br>(1.87)   | 0.01**<br>(2.06) |
| Economic Controls       | No                | Yes               | No                | Yes            | No                | Yes            | No                | Yes              |
| Census Tract FE         | Yes               | Yes               | Yes               | Yes            | Yes               | Yes            | Yes               | Yes              |
| Border $\times$ Year FE | Yes               | Yes               | Yes               | Yes            | Yes               | Yes            | Yes               | Yes              |
| Cluster (County)        | Yes               | Yes               | Yes               | Yes            | Yes               | Yes            | Yes               | Yes              |
| R <sup>2</sup> (Adj.)   | 0.257             | 0.259             | 0.340             | 0.339          | 0.399             | 0.394          | 0.379             | 0.373            |
| Observations            | 79920             | 77983             | 81114             | 79136          | 81634             | 79639          | 81337             | 79339            |

**Takeaway:** Approval ratio increased significantly among lowest income quartile consumers.

# Can Private Securitization Explain the Increase in Origination?

Was increased securitization behind increased origination?

|                         | Sold to Non-GSE   |                   | Sold to GSE       |                   | Not Sold          |                   |
|-------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                         | (1)<br>Fraction   | (2)<br>Fraction   | (3)<br>Fraction   | (4)<br>Fraction   | (5)<br>Fraction   | (6)<br>Fraction   |
| Treat $\times$ Post     | -0.007<br>(-0.54) | -0.006<br>(-0.48) | 0.039**<br>(2.22) | 0.042**<br>(2.33) | -0.004<br>(-0.58) | -0.001<br>(-0.12) |
| Economic Controls       | No                | Yes               | No                | Yes               | No                | Yes               |
| Census Tract FE         | Yes               | Yes               | Yes               | Yes               | Yes               | Yes               |
| Border $\times$ Year FE | Yes               | Yes               | Yes               | Yes               | Yes               | Yes               |
| Cluster (County)        | Yes               | Yes               | Yes               | Yes               | Yes               | Yes               |
| R <sup>2</sup> (Adj.)   | 0.003             | 0.002             | -0.000            | -0.000            | 0.032             | 0.034             |
| Observations            | 82667             | 80665             | 82667             | 80665             | 82667             | 80665             |

**Takeaway:** There is no significant increase in private securitization.

## Supply to Subprime Borrowers: Credit Score based Evidence

- Were the increased mortgages extended to subprime borrowers? Credit score-based evidence
- Prime consumer: Credit score  $\geq 620$

|                        | (1)                 | (2)                 | (3)               | (4)               |
|------------------------|---------------------|---------------------|-------------------|-------------------|
|                        | #Prime App.         | #Prime App.         | #Subprime App.    | #Subprime App.    |
| Treat $\times$ Post    | 308.58***<br>(3.39) | 312.51***<br>(3.33) | 10.48**<br>(2.12) | 10.78**<br>(2.16) |
| Economic Controls      | No                  | Yes                 | No                | Yes               |
| Zip3-State FE          | Yes                 | Yes                 | Yes               | Yes               |
| Border $\times$ Qtr FE | Yes                 | Yes                 | Yes               | Yes               |
| Cluster Zip3-State     | Yes                 | Yes                 | Yes               | Yes               |
| R <sup>2</sup> (Adj.)  | 0.757               | 0.758               | 0.792             | 0.792             |
| Observations           | 7711                | 7711                | 7711              | 7711              |

**Takeaway:** Increase in prime mortgage origination is more than the increase in subprime mortgages origination in the treated areas, relative to the control areas.



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